

Isabella Lamperti

PERSONAL DETAILS

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Website <https://isalamperti.github.io>
Languages: Italian (mother tongue), English (fluent), Spanish (conversational),
German (basic), French (basic)
Citizenship: Swiss, Italian
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SUMMARY

My research focuses on galaxy evolution and active galactic nuclei (AGN). In particular, I work on the study of the interstellar medium (ISM) of nearby star-forming galaxies and ultra-luminous infrared galaxies (ULIRGs), and on feedback effects from starburst and AGN winds. I have extensive experience working with multi-wavelength observational data, including optical and near-infrared spectroscopy, as well as submillimetre single-dish and interferometric observations.

EDUCATION & RESEARCH EXPERIENCE

Postdoctoral researcher, Centro de Astrobiología (CAB), CSIC-INTA, Spain 2026 - present
Project: *Dust properties of high-redshift galaxies*
Advisor: Dr. Irene Shivaiei

Postdoctoral researcher, Università di Firenze, Florence, Italy 2024 - 2026
Project: *Analysis of feedback in dual AGN and study of the ionised gas properties in nearby AGN using photoionization modelling*
Advisor: Prof. Alessandro Marconi

Postdoctoral researcher, Centro de Astrobiología (CAB), CSIC-INTA, Spain 2022 - 2024
Project: *Studying the spatially resolved properties of galaxies at cosmic noon using JWST/NIRSpec IFS, and developing new science cases and simulations for ELT/HARMONI*
Advisor: Prof. Santiago Arribas

Postdoctoral researcher, Centro de Astrobiología (CAB), CSIC-INTA, Spain 2020 - 2022
Project: *Investigating molecular gas outflows in ULIRGs using high-resolution ALMA observations*
Advisor: Dr. Miguel Pereira-Santaella

Ph.D. in Astrophysics, University College London (UCL), UK 2016 - 2020
Thesis Title: *Probing galaxy evolution through interstellar dust and gas properties*
Supervisors: Prof. Amélie Saintonge and Prof. Ilse de Looze

Ph.D. Studentship (1 year), European Southern Observatory, Germany 2019 - 2020
Project Title: *Integrated and spatially resolved dust properties of AGN hosts at $z \sim 2$*
Supervisors: Dr. Vincenzo Mainieri and Dr. Chris Harrison

Research Project, Swinburne University of Technology, Australia Aug.-Oct. 2016
Project Title: *Ages of massive star forming clumps in turbulent disks from H α equivalent widths in the DYNAMO survey*
Supervisors: Prof. Deanne Fisher and Prof. Karl Glazebrook

M.Sc. in Physics, ETH Zürich, Switzerland 2014-2016
Thesis Title: *Near-infrared spectroscopy of nearby hard X-ray selected AGN*
Supervisors: Dr. Michael Koss and Prof. Kevin Schawinski

B.Sc. in Physics, ETH Zürich, Switzerland 2010-2014
Bachelor Project Title: *Measurement of the accretion rate of nearby hard X-ray selected AGN using optical spectroscopy*
Supervisors: Dr. Michael Koss and Prof. Kevin Schawinski

SEMINARS, INVITED TALKS AND CONFERENCES

Dual and Binary AGN Workshop, ESTEC, NL, <i>Invited talk</i>	2026
European Southern Observatory in Vitacura, Santiago, Chile, <i>Colloquium</i>	2025
Osservatorio di Arcetri, Firenze, IT, <i>Informal talk</i>	2025
IFPU focus week, Trieste, IT, <i>Invited talk</i>	2024
Centro de Astrobiología (CAB), Madrid, ES, <i>Informal talk</i>	2023
Instituto de Astrofísica de Canarias (IAC), Tenerife, ES, <i>Invited talk</i>	2022
University College London, London, UK, <i>Department talk</i>	2019
University of Central Lancashire, Preston, UK, <i>Invited department seminar</i>	2018
University of Southampton, Southampton, UK, <i>Department talk</i>	2017
Durham University, Durham, UK, <i>Department talk</i>	2017
University of Cambridge, Cambridge, UK, <i>Informal talk</i>	2017
+11 contributed talks and 5 poster presentations at international conferences.	

AWARDS AND GRANTS

- Seal of Excellence: from the European Commission for the project proposal <i>Dust across cosmic times</i> , submitted under the Horizon 2020's Marie Skłodowska-Curie actions call	2022, 2023
- IFPU travel grant for the workshop <i>Dust in Trieste</i> (~400€)	2024
- European Southern Observatory (ESO) 1-year studentship grant (~20'000€)	2019-2020
- Royal Astronomical Society Travel Grant (~800€)	2018
- Swiss Society for Astrophysics and Astronomy (SSAA) Travel Grant (~900€)	2017
- Scholarship in Astronomy at the Swinburne University of Technology (~3000€)	2016

ACCEPTED PROPOSALS

Proposals as P.I.:

NOEMA: 45 hours, summer semester 2026 (co-P.I.: M.Perna).

The cold molecular gas content of dual AGN

ALMA: 14 hours, Cycle 11, 2024.

ISM at cosmic noon: star-formation and AGN activity resolved down to kpc-scales

ALMA: 8 hours, Cycle 9, 2022.

Tracing CO-dark gas in ULIRG outflows using [CI]

ALMA: 17 hours, Cycle 8, 2021.

Testing the SFE bimodality: measuring the CO-to-H₂ conversion factor in ULIRGs using ¹³CO

IRAM 30m: 21 hours, 2018.

Characterizing the millimetre emission in nearby galaxies with NIKA-2

Proposals as Co-I:

> 1000 hours at major facilities, including 80 hours at JWST, 100 hours at VLT, 300 hours of ALMA and ACA, 400 hours of APEX, 80 hours at VLA, and 70 hours at JCMT.

OBSERVING EXPERIENCE

VLT/ERIS: 2 nights observations as part of GTO programs (2025)

JCMT: 10 nights SCUBA-2 and RxA observations for the [JINGLE](#) survey (2017-2018)

IRAM 30m telescope: 7 nights CO spectroscopic observations for the [xCOLD GASS](#) survey (2017), 7 nights continuum observations with NIKA-2 (2018).

TEACHING

Marking of Physics exercises , University College London	2017-2019
Introduction to Astronomy, Atomic Physics	
Teaching Assistant at the Department of Mathematics , ETH Zürich	2014-2015
Mathematics I and II for architects	

STUDENTS SUPERVISION

- Yash Lapasia (first-year PhD student at CAB) since 2026
co-supervisor of his PhD Thesis
- Miguel Montero Vega (Master student at Universidad Complutense de Madrid) 2024
6-month internship at CAB, as primary advisor
- Dr. Lorenzo Ulivi (PhD student at the University of Firenze) 2023
visiting student at CAB for 3 months, as second advisor. [Published paper](#)

COMMUNITY SERVICES

- Referee for Nature Astronomy since 2025
- Referee for A&A and ApJ since 2023
- External reviewer for the Mexican TAC of the 10.4m GTC telescope 2026
- External reviewer for JWST proposals in Cycle 4 & 5 2024-2025
- Reviewer for ALMA proposals in Cycles 8-11,13 (distributed peer-review) 2021-2024
- Contribution to the white papers of the ALMA2040 AGN Working Group 2026
- Scientific organizer (co-chair) of the ENGAGE workshop at ESAC/Madrid (~ 30 participants) 2023
- Organizer of a one-day internal workshop of the ELT/HARMONI team at CAB (12 participants) 2024
- Starter and co-organizer of the Extragalactic Meetings (bi-weekly internal meetings) at CAB 2023-2024
- Co-organizer of the Lunch Talks of the Astrophysics group at UCL 2019

INTERNATIONAL COLLABORATIONS

I am involved in several international research programs that were awarded observing time on world's top telescopes. Among them:

- [BASS](#) (P.I. M. Koss): multiwavelength follow-up of AGN identified by the X-ray Swift/BAT observatory, using major facilities as VLT/X-shooter, VLT/MUSE, APEX, ALMA. I led the first data-release of the NIR spectra ([Lamperti et al. 2017](#)) and participated in 20 refereed publications.
- [JINGLE](#) (P.I. A. Saintonge): JCMT large program. I carried on the analysis of the dust properties using JCMT/SCUBA-2 and Herschel continuum data ([Lamperti et al. 2019](#)), and contributed to 3 additional publications.
- [SUPER](#) (P.I. V. Mainieri): ESO VLT/SINFONI large program with multiwavelength follow-up (including ALMA, JWST/MIRI). I was responsible for the analysis of the ALMA continuum data, published in [Lamperti et al. \(2021\)](#), and contributed to 4 additional publications, including one as third author ([Circosta, Mainieri, Lamperti et al. 2021](#)).
- [PUMA](#) (P.I. S. Arribas and M. Pereira-Santaella): MUSE and ALMA program. I was responsible for the analysis of the ALMA CO(2-1) data, published in [Lamperti et al. \(2022\)](#).
- [GA-NIFS](#) (P.I. S. Arribas and R. Maiolino): JWST/NIRSpec IFS GTO team. I am part of the team analysing the ALMA data available for the targets, contributed to the design of the website, and led the analysis of the galaxy GS5001 ([Lamperti et al. 2024](#)). I contributed to 21 additional publications, including one as third author ([Perna, Arribas, Lamperti et al. 2025](#)).
- [Nearby Galaxy Nuclei](#) (P.I. T. Böker): JWST/NIRSpec IFS GTO team. I contributed to the analysis of the data of the ULIRG Arp 220, which resulted in 2 refereed papers as 3rd author ([Perna, Arribas, Lamperti et al., 2024](#); [Ulivi, Perna, Lamperti et al., 2025](#)).
- [MIRACLE](#) (P.I. C. Marconcini, A. Feltre): JWST/MIRI MRS program to observe the nuclear region of seven nearby AGN. I led the analysis of the complementary ALMA CO data of the targets, which have been used to compare the molecular gas with the ionised gas phase. This project led to 3 refereed publications so far ([Marconcini, Feltre, Lamperti et al., 2025](#); [Ceci et al., 2026](#), [Marconcini et al., 2026](#)).
- [Cosmic Duets](#) (P.I. F. Mannucci, M. Scialpi): Survey aiming at confirming dual AGN at Cosmic Noon, using a data from a VLT/MUSE Large Programme and a VLT/ERIS GTO survey ([Scialpi et al., 2025](#); [Scialpi et al., 2026](#)). I have an accepted NOEMA proposal to obtain follow-up CO observations of the targets to constrain their molecular gas reservoir.

OUTREACH

- Interviewed by *New Scientist* for an [article](#) on recent research published by another team 2026
- Volunteer for school visits and outreach events at the Arcetri Observatory in Florence 2025
- Volunteer for the activity “Chat with an astronomer” organized by the Spanish Astronomical Society (SEA) for the International Day of Women and Girls in Science 2023
- Volunteer at the “Your Universe” event, a two-days exhibition for school pupils, organized annually at University College London 2020

SCHOOLS ATTENDED

- *A hands-on introduction to Machine Learning* (5-days course), Firenze, Italy 2025
- *Achieve your JWST data: getting ready for Cycle 4* (2-days workshop), Madrid, Spain 2024
- *European Radio Interferometry School (ERIS)* (5-days workshop), Gothenburg, Sweden 2019

MEMBERSHIPS OF SCIENTIFIC SOCIETIES

- Junior Member of the International Astronomical Union (IAU) since 2025
- Member of the European Astronomical Society (EAS) since 2017
- Member of the Swiss Society of Astronomy and Astrophysics (SSAA) since 2017
- Associated to the Italian National Institute of Astrophysics (INAF) since 2024

TECHNICAL SKILLS

Listed below are my key technical skills, pivotal in establishing international collaborations, and in generating a publication record of 92 refereed papers, including 7 as first author and 9 as 2nd or 3rd author (from [ADS library](#)).

- Analysis of integral field spectroscopic data coming from current major facilities working in the optical/near-infrared regime, including VLT/MUSE, VLT/ERIS and JWST/NIRSpec;
- Data reduction and analysis of sub-mm/mm data from ALMA and IRAM 30m.
- Spectral fitting for stellar continuum, emission and absorption lines in rest-frame ultra-violet, optical and near-infrared spectra, using standard routines (e.g. pPXF) as well as custom-developed analysis tools.
- Spectral energy distribution (SED) fitting of the dust emission in the far-infrared using modified black-body models.
- Application of Bayesian hierarchical methods to the analysis of galaxy samples.
- Physical and chemical characterisation of the ISM (e.g., ionisation, metallicity, electron density).
- Multi-phase (ionised, atomic, molecular) outflows characterisation through kinematic and physical analysis.